

S1 Appendix. Supplementary tables

A zero-shot foundation model versus city-trained models for 12-month dengue forecasting in eight Brazilian state capitals: A rolling-origin benchmark. Fabiano Novaes Barcellos Filho.

All tables are generated from the saved forecasts by `scripts/analyze_results.py` and `scripts/build_paper_assets.py` (<https://github.com/fabianofilho/dengue-timeseries-skforecast>, tag `paper-v2`). sMAPE, symmetric mean absolute percentage error; MAE, mean absolute error; RMSE, root mean squared error. Evaluation: expanding-window rolling origin, 121 origins per city (forecast windows starting January 2014 to January 2024), 12-month horizon.

Table A. sMAPE (%) with 95% moving block bootstrap intervals, primary model set.

City	TimesFM	CatBoost	XGBoost	RF	LightGBM	SARIMA	Prophet	Naive
São Paulo	78.0 (61.1–87.3)	75.6 (58.4–87.3)	81.2 (68.3–90.5)	77.0 (60.9–87.0)	88.9 (81.8–93.8)	87.8 (74.3–100.2)	77.3 (60.0–87.8)	77.6 (63.8–85.1)
Rio de Janeiro	96.8 (84.8–109.6)	97.5 (77.6–113.7)	101.5 (89.0–112.8)	93.5 (75.9–105.9)	101.2 (84.8–112.3)	101.7 (87.8–111.9)	109.6 (85.2–131.1)	104.1 (89.3–119.0)
Belo Horizonte	97.1 (85.7–111.0)	62.0 (45.4–75.8)	76.4 (58.0–90.8)	70.3 (52.0–85.8)	92.1 (81.5–102.5)	95.8 (87.8–107.7)	80.7 (68.4–90.9)	79.6 (69.6–89.5)
Brasília	74.3 (62.8–86.4)	76.6 (67.1–84.0)	90.3 (76.5–101.1)	81.5 (68.8–90.9)	84.2 (74.5–90.1)	80.2 (70.7–89.3)	79.3 (63.8–92.8)	74.2 (64.5–83.9)
Fortaleza	64.2 (56.6–75.0)	69.4 (58.6–86.6)	79.7 (69.1–93.5)	68.4 (56.8–84.0)	73.7 (66.5–85.7)	73.4 (65.3–83.8)	69.7 (58.9–86.6)	69.6 (59.5–86.8)
Recife	67.3 (56.7–78.1)	65.9 (50.3–80.5)	73.4 (58.7–86.7)	71.0 (57.4–82.7)	77.5 (65.6–88.9)	87.1 (75.1–96.6)	81.5 (61.5–103.4)	83.3 (67.6–101.0)
Manaus	56.0 (47.3–61.9)	55.7 (47.6–61.6)	54.8 (44.3–61.6)	53.9 (45.5–59.4)	62.8 (54.7–69.3)	62.4 (50.3–71.5)	53.8 (41.3–59.4)	59.1 (46.7–68.4)
Salvador	68.9 (59.5–82.3)	71.8 (56.7–86.5)	72.2 (57.6–84.0)	71.0 (57.2–84.9)	72.3 (63.6–82.1)	79.5 (62.6–97.4)	88.9 (68.4–108.0)	87.2 (64.7–112.3)
Mean	75.3	71.8	78.7	73.3	81.6	83.5	80.1	79.3

Trained models were fitted on log1p-transformed counts; TimesFM was used zero-shot on raw counts. RF, Random Forest; Naive, seasonal naive. Bold, lowest value in the row.

Table B. MAE (cases per month), primary model set.

City	TimesFM	CatBoost	XGBoost	RF	LightGBM	SARIMA	Prophet	Naive
São Paulo	9,617	9,611	9,715	9,629	9,956	11,888	9,797	9,938
Rio de Janeiro	1,966	2,366	3,349	2,282	2,753	3,958	2,071	2,316
Belo Horizonte	6,384	5,707	6,578	6,193	7,499	16,192	6,431	8,294
Brasília	3,823	3,965	4,290	4,090	4,095	5,558	4,023	4,319
Fortaleza	1,434	1,596	2,093	1,593	1,825	2,273	1,624	1,698
Recife	737	841	919	927	979	2,201	1,066	1,141
Manaus	269	270	301	257	299	301	261	298
Salvador	423	453	460	440	451	834	559	601
Mean	3,082	3,101	3,463	3,176	3,482	5,401	3,229	3,576

Trained models were fitted on log1p-transformed counts; TimesFM was used zero-shot on raw counts. RF, Random Forest; Naive, seasonal naive. Bold, lowest value in the row.

Table C. RMSE (cases per month), primary model set.

City	TimesFM	CatBoost	XGBoost	RF	LightGBM	SARIMA	Prophet	Naive
São Paulo	44,024	44,152	44,232	44,136	43,519	43,563	43,486	43,114
Rio de Janeiro	5,958	6,310	8,380	6,395	6,022	14,873	6,310	6,285
Belo Horizonte	17,388	17,620	17,786	17,586	17,015	91,580	17,548	20,006
Brasília	12,669	12,786	13,165	12,973	12,755	14,695	11,658	12,870
Fortaleza	2,693	2,841	3,532	2,865	3,066	5,451	2,777	3,083
Recife	1,474	1,616	1,731	1,666	1,685	10,907	1,984	2,005
Manaus	526	520	633	508	519	561	526	570
Salvador	713	772	778	729	730	2,039	885	930
Mean	10,681	10,827	11,280	10,857	10,664	22,959	10,647	11,108

Trained models were fitted on log1p-transformed counts; TimesFM was used zero-shot on raw counts. RF, Random Forest; Naive, seasonal naive. Bold, lowest value in the row.

Table D. sMAPE (%) with all trained models fitted on untransformed counts (sensitivity analysis).

City	TimesFM	CatBoost	XGBoost	RF	LightGBM	SARIMA	Prophet	Naive
São Paulo	78.0	78.7	84.8	87.3	119.8	87.1	89.7	77.6
Rio de Janeiro	96.8	126.4	119.7	133.3	154.7	135.6	171.1	104.1
Belo Horizonte	97.1	95.2	88.7	90.2	151.4	134.0	129.0	79.6
Brasília	74.3	76.6	83.7	82.4	100.5	103.2	78.2	74.2
Fortaleza	64.2	77.5	89.4	98.4	102.1	87.6	79.2	69.6
Recife	67.3	83.3	83.3	81.2	111.2	106.1	102.7	83.3
Manaus	56.0	70.9	65.5	67.9	94.8	99.7	173.6	59.1
Salvador	68.9	76.0	72.1	73.7	82.2	91.0	99.5	87.2
Mean	75.3	85.6	85.9	89.3	114.6	105.5	115.4	79.3

Same design as the primary analysis, without the log1p transformation. Bold, lowest value in the row.

Table E. sMAPE (%) for the 109 origins whose forecast windows ended by December 2023 (sensitivity analysis).

City	TimesFM	CatBoost	XGBoost	RF	LightGBM	SARIMA	Prophet	Naive
São Paulo	71.7	69.4	75.5	71.5	87.9	86.3	70.6	72.7
Rio de Janeiro	96.0	95.9	101.3	91.2	101.3	102.8	103.0	102.8
Belo Horizonte	95.5	57.1	72.7	66.5	89.7	97.1	76.0	77.7
Brasília	72.4	75.4	88.8	80.2	82.7	78.9	79.3	72.6
Fortaleza	66.5	71.9	79.7	68.3	74.7	74.2	71.8	71.9
Recife	66.7	65.3	72.4	69.8	77.0	86.6	82.7	84.0
Manaus	54.1	54.7	53.0	51.9	62.0	59.8	49.8	56.6
Salvador	69.5	70.8	71.3	72.1	72.1	78.8	84.9	87.4
Mean	74.0	70.1	76.8	71.5	80.9	83.1	77.3	78.2

Trained models were fitted on log1p-transformed counts; TimesFM was used zero-shot on raw counts. RF, Random Forest; Naive, seasonal naive. Bold, lowest value in the row.

Table F. TimesFM versus each comparator: paired sMAPE difference and Diebold-Mariano test. Difference = TimesFM minus comparator sMAPE, in percentage points (negative favors TimesFM), with 95% paired moving block bootstrap interval over forecast origins (blocks of 12, 2,000 replicates). DM, Diebold-Mariano statistic on origin-level sMAPE losses with Newey-West variance (12 lags) and Harvey-Leybourne-Newbold correction; p adjusted with Holm across all 56 comparisons. RF, Random Forest; Naive, seasonal naive.

City	Comparator	Difference (95% CI)	DM	p (Holm)
São Paulo	CatBoost	+2.4 (−10.2 to 17.5)	0.31	1.000
São Paulo	XGBoost	−3.2 (−14.3 to 8.3)	−0.50	1.000
São Paulo	RF	+1.0 (−11.3 to 15.7)	0.13	1.000
São Paulo	LightGBM	−10.9 (−28.5 to 0.6)	−1.25	1.000
São Paulo	SARIMA	−9.8 (−28.6 to 3.0)	−1.05	1.000
São Paulo	Prophet	+0.7 (−5.1 to 6.7)	0.22	1.000
São Paulo	Naive	+0.4 (−14.9 to 13.6)	0.05	1.000
Rio de Janeiro	CatBoost	−0.7 (−14.4 to 15.2)	−0.08	1.000
Rio de Janeiro	XGBoost	−4.7 (−17.2 to 7.4)	−0.68	1.000
Rio de Janeiro	RF	+3.3 (−7.4 to 16.5)	0.49	1.000
Rio de Janeiro	LightGBM	−4.4 (−13.4 to 7.5)	−0.72	1.000
Rio de Janeiro	SARIMA	−4.9 (−17.5 to 12.4)	−0.60	1.000
Rio de Janeiro	Prophet	−12.8 (−31.6 to 8.8)	−1.09	1.000
Rio de Janeiro	Naive	−7.3 (−21.0 to 5.1)	−1.02	1.000
Belo Horizonte	CatBoost	+35.1 (19.9 to 56.9)	3.38	0.055
Belo Horizonte	XGBoost	+20.6 (6.4 to 42.9)	2.01	1.000
Belo Horizonte	RF	+26.8 (11.6 to 48.4)	2.59	0.565
Belo Horizonte	LightGBM	+4.9 (−11.3 to 26.5)	0.49	1.000
Belo Horizonte	SARIMA	+1.3 (−16.3 to 18.5)	0.13	1.000
Belo Horizonte	Prophet	+16.3 (4.2 to 35.4)	1.87	1.000
Belo Horizonte	Naive	+17.5 (2.6 to 36.0)	1.86	1.000
Brasília	CatBoost	−2.4 (−10.9 to 8.7)	−0.43	1.000
Brasília	XGBoost	−16.0 (−28.7 to −3.0)	−2.29	1.000
Brasília	RF	−7.3 (−16.3 to 3.3)	−1.29	1.000
Brasília	LightGBM	−9.9 (−20.2 to 4.4)	−1.46	1.000
Brasília	SARIMA	−6.0 (−19.7 to 9.0)	−0.76	1.000
Brasília	Prophet	−5.0 (−17.0 to 13.2)	−0.56	1.000
Brasília	Naive	+0.1 (−8.5 to 10.6)	0.02	1.000
Fortaleza	CatBoost	−5.2 (−16.5 to 5.1)	−0.89	1.000
Fortaleza	XGBoost	−15.5 (−28.6 to −3.9)	−2.29	1.000
Fortaleza	RF	−4.2 (−16.4 to 7.2)	−0.65	1.000
Fortaleza	LightGBM	−9.5 (−21.2 to 1.9)	−1.55	1.000
Fortaleza	SARIMA	−9.2 (−19.7 to 1.4)	−1.63	1.000
Fortaleza	Prophet	−5.5 (−16.7 to 5.1)	−0.90	1.000
Fortaleza	Naive	−5.4 (−18.0 to 5.7)	−0.84	1.000
Recife	CatBoost	+1.4 (−8.7 to 13.9)	0.22	1.000
Recife	XGBoost	−6.1 (−13.4 to 3.1)	−1.33	1.000
Recife	RF	−3.8 (−11.9 to 5.6)	−0.82	1.000
Recife	LightGBM	−10.2 (−20.4 to −0.4)	−1.88	1.000
Recife	SARIMA	−19.9 (−26.9 to −11.2)	−4.30	0.002
Recife	Prophet	−14.2 (−30.3 to −0.3)	−1.70	1.000
Recife	Naive	−16.1 (−28.5 to −6.8)	−2.71	0.404
Manaus	CatBoost	+0.3 (−4.9 to 4.6)	0.12	1.000
Manaus	XGBoost	+1.3 (−6.1 to 7.9)	0.33	1.000
Manaus	RF	+2.2 (−2.5 to 6.9)	0.84	1.000
Manaus	LightGBM	−6.8 (−17.8 to 2.4)	−1.20	1.000
Manaus	SARIMA	−6.3 (−15.5 to 1.3)	−1.37	1.000
Manaus	Prophet	+2.2 (−2.4 to 10.6)	0.57	1.000
Manaus	Naive	−3.0 (−14.5 to 6.5)	−0.51	1.000

City	Comparator	Difference (95% CI)	DM	p (Holm)
Salvador	CatBoost	-3.0 (-11.7 to 9.3)	-0.50	1.000
Salvador	XGBoost	-3.3 (-10.3 to 8.2)	-0.57	1.000
Salvador	RF	-2.1 (-9.9 to 7.4)	-0.41	1.000
Salvador	LightGBM	-3.5 (-7.8 to 5.0)	-0.92	1.000
Salvador	SARIMA	-10.6 (-21.0 to 2.0)	-1.74	1.000
Salvador	Prophet	-20.1 (-31.1 to -5.8)	-2.73	0.393
Salvador	Naive	-18.3 (-32.6 to -3.2)	-2.21	1.000